ISRA UNIVERSITY HYDERABAD
DEPARTMENT OF COMPUTER SCIENCE
FINAL YEAR PROJECT REPORT

PetAdopt

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Certificate

This is to certify that Rajwanti, Ghufrana Abbasi, Paras Abbasi, and Jaweria have worked under my supervision for the BS(CS/IT) project titled PetAdopt, as per the partial requirement for the award of the degree of BS(CS/IT), and the work is original and satisfactory.

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We take pride in the successful completion of PetAdopt and look forward to the positive impact it may bring in the future.

ABSTRACT

PetAdopt is a mobile application designed to transform how pet adoption and care services are accessed by providing a unified platform for users and service providers. This project was created to address the common challenges of inefficiency, lack of transparency, and disconnected services in traditional pet adoption systems. PetAdopt serves four key user roles: Adopter, Shelter/Owner, Service Provider, and Admin, each supported by dedicated interfaces and real time communication features. Adopters can discover pets, submit adoption requests, and book grooming services, while shelters manage listings, appointments, and approvals through a structured backend. Admins ensure platform integrity through verification and oversight.

The application was developed using Flutter for cross platform compatibility, with backend services powered by Node.js and MongoDB. User centred design principles guided its development, informed by comprehensive surveys and interviews focusing on local adoption practices and user trust concerns. Application testing, including unit testing, system testing, and user acceptance testing, confirmed the application's reliability, responsiveness, and usability. Evaluation results demonstrated high user satisfaction, particularly with the platform's transparent adoption tracking, integrated service booking, verified profile system, and secure payment processing functionality.

PetAdopt addresses a significant gap in the pet care landscape by offering an accessible, efficient, and secure digital solution. It empowers users with convenience and choice while enabling shelters and service providers to expand their reach and operational capabilities. The thoughtful integration of technology with user centric design principles makes PetAdopt a scalable solution positioned for substantial nationwide impact in the animal welfare sector, with potential for future expansion into additional pet care services.

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CHAPTER 1
INTRODUCTION

1.1. PROJECT BACKGROUND

In the past, finding and adopting a pet was often frustrating and complicated. People had to visit multiple animal shelters or search through unreliable online listings. This process was time consuming, and it was hard to know if the information about pets was accurate or trustworthy. Many potential adopters gave up because the system felt disorganized and unclear. At the same time, animal shelters struggled to manage the high number of inquiries, schedule visits, and keep their online profiles updated. This often meant that great pets were overlooked simply because they weren't visible or easy to connect with.

Seeing these challenges inspired us to create **"PetAdopt"**, a mobile app that makes pet adoption simpler, faster, and more trustworthy. Instead of dealing with outdated and confusing processes, users can now browse verified pets, chat directly with shelters, and complete adoption steps right from their phone.

PetAdopt brings together four main groups: people looking to adopt, shelters and rescue groups, Service providers and platform administrators. Shelters sign up and are verified to make sure they're legitimate. Adopters can search for pets, submit applications, and schedule meetings in just a few taps. Service providers (grooming services) can give services to pet owners and Admins help keep the platform safe, reliable, and easy to use for everyone.

Our goal is to help more pets find loving homes by making adoption accessible, transparent, and stress free. With so many people using smartphones today, PetAdopt meets a real need creating a friendly and practical solution for pet adoption in the digital age.

1.2. Problem Statement

The process of adopting a pet, as experienced by most people today, remains complicated and inefficient. Despite digital advancements in other areas, the pet adoption sector has been slow to embrace technology, especially in developing regions.

Pet Adopters face several challenges:

- They have to visit multiple shelters physically, which takes time and effort.
- There is often no clear information about which pets are actually available.
- Communication with shelters is often slow and unreliable.
- Many potential adopters feel uncertain about the entire process.

Animal shelters and individual pet owners:

- They struggle to manage numerous inquiries and appointment requests.
- There is a limited digital presence to reach potential adopters effectively.
- Manual record keeping leads to errors and confusion.
- It's challenging to build trust with people who want to adopt
- Good animals often remain unnoticed due to poor visibility.

Pet care services Providers:

- There is no centralised platform to find verified groomers, trainers, or veterinarians.
- Service providers struggle to reach potential customers effectively.
- Quality and pricing information is often unclear or unavailable.
- New pet owners lack guidance on essential post adoption services.

This situation clearly demands a platform that can make pet adoption more organised, transparent, and trustworthy for everyone involved.

1.3. Proposed Solution

PetAdopt addresses all the above challenges by acting as a digital bridge between pet adopters, animal shelters, and service providers. The solution includes:

- A mobile application with seperate interfaces for adopters, shelters, and service providers.
- A secure registration system where shelters and service providers can register.
- An intuitive booking system allowing users to browse pets, submit applications, and schedule visits.
- Service booking functionality for grooming, veterinary care, and other pet services.
- Integrated communication tools for direct messaging between all parties

The app helps shelters and service providers expand their reach while enabling adopters to find pets and access care services conveniently from their homes.

1.4. Objectives

The primary objectives of PetAdopt are:

- To simplify the pet adoption process through an easy-to-use mobile application
- To provide a platform for shelters and service providers to manage their operations efficiently
- To reduce the time pets spend in shelters through better visibility and streamlined processes
- To increase accessibility for people who cannot frequently visit physical locations
-

1.5. Characteristics of the proposed Solution

PetAdopt stands out due to its user-centred features:

- **Verification and Trust:** All shelters and service providers undergo document verification to ensure legitimacy.
- **Unified Platform**: Combines adoption services with essential pet care in one application
- **Real-Time Management:** Instant updates and notifications keep all users informed
- **Role-Based Design:** Separate interfaces optimised for adopters, shelters, service providers, and admins
- **Accessibility Focus:** Simple design ensures ease of use for all technical skill levels
- **Scalable Architecture:** Built to accommodate future expansions and additional features
- **Transparent Reviews:** Rating system helps users make informed decisions and maintains quality standards

1.6. Scope of the project

The **PetAdopt** application is developed as a cross platform mobile app using Flutter, which can be extended to web platforms in the future. The initial focus is on urban and semi-urban areas where pet adoption awareness is growing and smartphone usage is widespread.

Pet Owner Features:

- Account and profile management
- Listing pets for adoption with photos and details
- Managing pet availability status
- Communication with potential adopters
- Scheduling meetings with interested adopters
- Reviewing adoption applications

Adopter Features:

- User registration and profile creation
- Browsing available pets with search and filter options
- Viewing detailed pet profiles with complete information
- Submitting adoption applications online
- Scheduling visits with pet owners/shelters
- In app messaging with pet owners/shelters
- Rating and review system after adoption

Service Provider Features:

- Service listing and pricing management
- Appointment scheduling
- Booking management for grooming/veterinary services
- Client communication through messaging
- Rating and review visibility
- Verification and approval of all user registrations

The application will initially support the core adoption process and basic services, with plans to expand into additional pet care services such as pet insurance and a pet supplies marketplace in future versions.

1.7. Outline of Report

The next chapter presents a literature review examining existing pet adoption systems and their limitations. Chapter 3 discusses the methodology used in developing PetAdopt, followed by implementation details in Chapter 4. Finally, Chapter 5 presents the conclusions and recommendations for future improvements to the platform.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

This chapter presents an overview of the current state of traditional and digital pet adoption methods, reviews existing mobile platforms for pet adoption and care, highlights their limitations in Pakistan's context, and identifies the gaps that PetAdopt aims to address.

In today's digital age, technology has become deeply integrated into every aspect of human life, from transportation and healthcare to education and entertainment. However, one area that has remained relatively underdeveloped, especially in developing regions, is pet adoption and animal welfare management. Traditional pet adoption systems often rely on outdated methods that lack transparency, coordination, and accessibility, making it difficult for both adopters and shelters to connect efficiently.

In the past, individuals who wanted to adopt pets had to visit multiple shelters or rely on informal social media listings to find available animals. This process was not only time consuming but also unreliable, as information was often outdated or incomplete. Likewise, animal shelters faced challenges managing numerous inquiries, handling paperwork, and maintaining updated profiles of pets. These inefficiencies led to missed adoption opportunities and longer shelter stays for many animals.

With the increasing adoption of smartphones, internet access, and digital services, the idea of a centralised online platform for pet adoption and care has gained significant attention. Just as technology has simplified the process of booking cabs or ordering food, it holds the potential to simplify and transform how people adopt and care for pets. A well designed mobile application can bring adopters, shelters, and service providers together under one ecosystem, ensuring smooth communication, transparent processes, and verified information.

This changing behaviour among users seeking convenience, trust, and real time updates has created the foundation for platforms like PetAdopt. Unlike traditional methods, PetAdopt uses modern technology to connect verified shelters, service providers, and potential adopters in one safe, user friendly environment. The platform not only promotes adoption but also integrates post-adoption care services such as grooming, training, and veterinary support, making it a complete pet care ecosystem.

2.2 Limitations of Traditional Pet Adoption Methods

2.2.1 Manual and Disconnected Processes

Conventional pet adoption systems rely heavily on manual communication, such as phone calls, physical visits, or paper based applications. This makes the process slow and error prone. Adopters often have to travel to multiple shelters just to inquire about pet availability. Shelters, on the other hand, face difficulties maintaining organised records, leading to confusion, missed follow ups, or double bookings. These outdated processes make the overall experience frustrating for both parties.

2.2.2 Limited Accessibility and Awareness

In most areas, shelters operate within limited working hours and have minimal online presence. People living in semi urban or remote regions find it particularly hard to access shelter information or visit in person. Moreover, lack of awareness about adoption procedures or nearby shelters reduces the chances of successful adoptions.

2.2.3 Poor Communication Channels

Communication between adopters and shelters is often slow and inconsistent. Adopters may wait days to receive responses to basic inquiries, while shelters struggle to manage a high volume of messages and calls. Without a real time messaging, misunderstandings and delays are common, often leading to missed adoption opportunities.

2.2.3 No Centralised Feedback or Tracking System

Traditional systems rarely include structured feedback or tracking mechanisms. Once a pet is adopted, there is little follow up or record of the adoption history. Adopters cannot easily share their experiences, and shelters miss valuable insights that could improve their services. Similarly, there is no central database to monitor adoption rates or identify trends in pet welfare.

2.2.4 Difficulty in Connecting with Post Adoption Services

After adopting a pet, owners often need grooming, veterinary, or training services. However, these services are usually scattered, and there is no integrated platform where users can book them conveniently. This gap makes it hard for adopters to find trusted providers and maintain proper pet care after adoption.

In summary, the traditional pet adoption process is fragmented, slow, and unreliable. It lacks modern solutions for trust, convenience, and accessibility. These challenges underline the need for a comprehensive platform like PetAdopt, which integrates communication, and real time tracking into one simple mobile application.

Over the years, pet adoption has largely been managed through informal networks, word of mouth, or physical visits to animal shelters. These traditional methods often result in limited visibility for adoptable pets, inefficient communication between adopters and shelter. Additionally, many potential adopters in Pakistan face difficulties in finding accurate information about available pets, adoption requirements, and post adoption care services.

While some countries have developed digital platforms that connect adopters with shelters or service providers, such solutions remain limited or underdeveloped in Pakistan. Existing online listings, such as social media groups or small scale websites, often lack structured data management and secure communication channels. This creates trust issues and inefficiencies that discourage people from adopting through digital means.

Recognising these challenges, PetAdopt aims to provide a comprehensive digital solution that brings together adopters, shelters, and service providers on a single platform. The system ensures transparent communication and easy access to services such as grooming and veterinary care. By addressing the current gaps in accessibility, trust, and organisation, PetAdopt seeks to modernise the pet adoption ecosystem in Pakistan and promote responsible pet ownership through technology.

2.3 Rise of Digital Pet Adoption Platforms

2.3.1 Global Shift Toward Online Pet Adoption

In recent years, the process of adopting pets has undergone a major transformation due to the increasing integration of digital technologies. Just as on demand platforms revolutionised transportation, food delivery, and home services, mobile applications are now reshaping how people find and adopt pets. Instead of relying on physical shelter visits or newspaper listings, potential adopters can now browse available pets online, read verified details, and connect directly with shelters through apps or websites.

This shift has been particularly evident in countries with strong digital infrastructure, where people value convenience, transparency, and time efficiency. The COVID-19 pandemic further accelerated the adoption of online solutions, as many shelters and individuals moved toward virtual adoption processes for safety and accessibility. As a result, several platforms such as Petfinder (USA), Adopt a Pet (USA), and Pets4Homes (UK) have emerged as trusted digital intermediaries connecting millions of adopters and animals globally.

2.3.2 Features of Leading Pet Adoption Platforms

Modern pet adoption platforms are designed to simplify the process and ensure safety for both adopters and pets. Common features include:

- Detailed pet profiles with photos, health records, and temperament details
- Search and filter options by breed, age, and location.
- Verified shelters and foster networks

- In app communication tools for transparency

While these features represent a significant improvement over traditional methods, many of these platforms cater mainly to developed countries. They rely heavily on advanced digital payment systems, wide adoption of microchipping, and active animal welfare databases systems that are not yet standardised in developing regions like Pakistan.

2.3.3 Gaps in Current Offerings

Despite their global success, existing pet adoption platforms face several limitations when applied in Pakistan's context. Many lack local language support and offline accessibility suited for regional needs. Furthermore, most platforms focus solely on facilitating adoption and do not provide post adoption support services, such as veterinary care, grooming, or training assistance.

Another major gap lies adopters often struggle to find legitimate service providers for their pets after adoption, leading to fragmented care experiences.

2.3.4 The Regional Perspective: Pakistan's Untapped Market

Pakistan's pet industry is growing steadily, with rising awareness about animal welfare and an increasing number of pet lovers, especially in urban areas. However, the adoption process remains largely informal, dependent on word of mouth, social media groups, or small independent shelters. These channels, while helpful, lack structure, and a centralised system to manage adoption requests or track pet welfare.

Moreover, the country's digital landscape presents unique challenges. Many shelters lack access to digital tools or training, and digital payment options are not universally adopted. As a result, most adoption related interactions still happen manually, leading to inefficiency and limited reach. Only a few local initiatives have attempted to digitalise this process, but they often fail due to poor user trust, lack of verification systems, and minimal post adoption support.

2.3.5 How PetAdopt Fits the Landscape

PetAdopt was developed specifically to address these regional gaps and modernise the pet adoption ecosystem in Pakistan. The platform introduces:

- A role based architecture (Admin, Adopter, Shelter, Service Provider) for transparency and accountability
- Offline friendly features to accommodate users without digital payment access
- Post adoption service integration (vet, grooming, and training) for long term pet care
- Admin monitoring and complaint management to ensure a safe, reliable environment

Unlike global platforms that primarily focus on convenience, PetAdopt emphasises trust, inclusivity, and community impact. It bridges the gap between traditional adoption practices and modern digital accessibility, empowering shelters, simplifying adoption for users, and ensuring pets find responsible homes.

The global success of digital pet adoption systems demonstrates that technological transformation is both possible and necessary for animal welfare. However, localised solutions like PetAdopt are essential to adapt these global standards to the social, cultural, and technological realities of countries like Pakistan.

2.4 Comparative Analysis of Existing Systems

To understand the position of PetAdopt within the broader digital adoption ecosystem, it is essential to compare its design and functionality with existing global and regional platforms. This comparison highlights the differences in scope, accessibility, and user focus, demonstrating how PetAdopt is uniquely tailored to Pakistan's context.

2.4.1 Overview of Selected Platforms

The following table provides a comparative summary of well known pet adoption platforms and their key features.

Platform	Region	Core Features	Limitations	Relevance to Pakistan
Petfinder	USA, Canada	Extensive pet database; adoption forms; advanced filters	Limited to North America; relies on established shelter networks	regional restrictions and payment dependencies
Adopt-a-Pet	USA	Partnered shelters; charity integration	Requires a strong animal registry system; not localised for developing countries	Lacks adaptability to local systems
Pets4Homes	UK, EU	Private and shelter listings; user ratings; in-app messaging	Lacks adoption tracking and verification depth	Moderate useful model, but lacks localised verification
OLX Pets Section	Global (including Pakistan)	Free listings; wide reach; accessible to all users	No verification; risk of fraudulent listings; no adoption tracking	popular but unreliable due to a lack of regulation
Local Facebook Groups	Pakistan	Community-based adoption posts; social interaction	Unverified information; no data tracking or post-adoption support	commonly used but lacks formal structure
PetAdopt *(Proposed)*	Pakistan	Verified users; role-based access; adoption tracking; in-app chat; service provider integration; admin oversight	Currently under initial deployment	specifically designed for Pakistan's needs

Table 2.1 Selected platforms

2.4.2 Analysis of Existing Platforms

The analysis reveals that while global platforms such as Petfinder, Adopt-a-Pet, and Pets4Homes have advanced adoption management systems, their success depends on robust digital infrastructures,, and nationwide animal welfare databases. These prerequisites are currently underdeveloped in Pakistan, making direct implementation impractical.

On the other hand, local alternatives such as OLX and Facebook groups are widely used due to accessibility and familiarity but lack structured management, adoption monitoring, and secure communication channels. These gaps lead to misinformation, low trust, and potential misuse of adoption platforms.

PetAdopt bridges this divide by combining the technological strengths of global systems with the local adaptability required in Pakistan. Its user roles approach ensures only legitimate shelters and service providers operate on the platform, while its real time communication features make it inclusive for users. Additionally, by integrating post adoption services (grooming, veterinary care, and training), PetAdopt goes beyond existing models to promote responsible and long-term pet care.

2.4.3 Summary

The comparative analysis highlights that no current platform fully meets the combined needs of transparency and localised accessibility for the Pakistani market. PetAdopt stands out as a comprehensive solution that aligns global best practices with national realities. Its hybrid model blending digital innovation, trust mechanisms, and inclusive service design positions it as a scalable framework for modernising the pet adoption landscape in Pakistan.

2.5 Value of Trust, Safety, and Personalisation

In any service that connects individuals for a sensitive and emotional purpose, such as adopting or rehoming a pet, three values stand out as fundamental: **trust, safety, and personalisation**. These values are not optional; they are essential for ensuring a positive and reliable adoption experience. For a digital platform like **PetAdopt**, which bridges the gap between adopters, pet owners, and shelters, these values are the foundation for building user confidence and long-term engagement.

2.5.1 Trust: Building Confidence in a Digital Environment

Trust is the cornerstone of every adoption process. Both adopters and pet owners must feel assured that the people they are interacting with have genuine intentions and care for animal welfare. Adopters need confidence that the pets listed are real, healthy, and accurately described, while pet owners and shelters must trust that the adopters will provide safe, loving homes.

PetAdopt fosters trust through multiple layers:

- **Ratings and Reviews:** Adopters can rate their experience with shelters or individual users, helping future users make informed decisions.
- **Profile Transparency:** User profiles clearly display name, photo, city, and adoption history to build credibility and accountability.

2.5.2 Safety: Protecting Users and Pets Alike

The process of adopting or rehoming a pet involves emotional and physical safety for both humans and animals. **PetAdopt's** safety model emphasises prevention, monitoring, and quick response to issues.

- **Secure Communication:** The in app chat system allows adopters and pet owners to communicate safely without sharing personal contact details.
- **Health and Welfare Checks:** Shelters and users are encouraged to upload veterinary health records to ensure the pet well being before adoption.

2.5.3 Personalisation: A Tailored Experience for Every User

Every adopter has unique preferences; some may prefer small dogs, others may want cats, birds. Personalisation ensures that users find pets that truly match their lifestyle. PetAdopt integrates personalisation through:

- **Category and Filter Options:** Users can filter pets by species, breed, age, gender, vaccination status, and location.

2.5.4 Balancing the Three Values

Finding the right balance between trust, safety, and personalisation is crucial for user satisfaction and system integrity. Overregulation can make adoption slow or complicated, while too much flexibility can invite fraud or misuse. PetAdopt maintains a balanced approach:

- **Open Feedback:** Builds credibility and community confidence.
- **Personalized Filters + Secure Communication:** Offers a user friendly experience without compromising security.

Through this balanced system, PetAdopt ensures reliability, user engagement, and a compassionate adoption process.

2.5.5 The Cultural Context of Pakistan

In Pakistan, pet adoption carries unique social and cultural sensitivities. Many people prefer adopting from trusted individuals or local shelters due to fear of scams or health concerns. PetAdopt addresses these challenges by incorporating:

- **Localised Language Support:** The platform supports Urdu and English for accessibility.
- **Community Trust Features:** regional listings help users connect with trusted local adopters and shelters.

Ultimately, trust, safety, and personalisation are deeply interconnected values that determine the success of any digital adoption platform. By integrating these principles into its core design and functionality, PetAdopt goes beyond a simple listing app; it builds a community of responsible, caring, and trustworthy adopters and pet owners. This approach nurtures loyalty, compassion, and peace of mind for all users involved.

2.6 Addressing the Gaps in Existing Systems

Although several global and regional platforms have introduced digital solutions for pet adoption, few have addressed the specific social, technical, and cultural challenges present in developing countries like Pakistan. Existing systems often fail due to gaps in accessibility, trust, and inclusivity. This section identifies the main shortcomings of current platforms and outlines how **PetAdopt** effectively addresses each one.

2.6.1 Gaps

1. Lack of Offline Accessibility and Payment Flexibility

Most international platforms rely entirely on online systems requiring digital payments, and continuous internet connectivity. However, in Pakistan, many users lack access to digital wallets or prefer cash based transactions. This restricts participation, especially in semi-urban and rural areas.

PetAdopt's Response:

PetAdopt is designed to operate inclusively. Users can manage adoption processes and communication through the app while completing transactions (such as adoption fees or donations) offline. This hybrid model ensures participation from users across all income and digital literacy levels.

2. Urban Centric Digital Adoption

Most pet adoption platforms are concentrated in major cities with developed infrastructure. This leaves smaller cities, towns, and rural communities largely excluded from the digital adoption ecosystem.

PetAdopt's Response:

PetAdopt's location based filter matching and flexible regional filters allow users to discover pets and shelters within their region

3. Cultural and Ethical Sensitivity Gaps

Global apps often overlook cultural norms, local adoption ethics, and community attitudes toward pets. In Pakistan, adoption decisions are influenced by religious beliefs, space availability, and social perception.

PetAdopt's Response:

PetAdopt respects cultural sensitivities by offering community-focused adoption campaigns, education on responsible pet ownership, and localised service listings (such as small pets or rescue animals). It also collaborates with local NGOs to promote ethical adoption practices instead of commercial breeding.

While global solutions provide a valuable starting point, they frequently ignore the local realities of accessibility, culture, and user trust. PetAdopt is purpose built to adapt digital pet adoption to Pakistan's social and technological context. By systematically identifying and bridging these gaps, PetAdopt positions itself as an inclusive, reliable, and community driven platform that empowers adopters, shelters, and pet owners alike.

CHAPTER # 3

SURVEY ANALYSIS

3.1. Introduction

To better understand user behaviour, needs, and expectations related to pet adoption and care services, an online survey was conducted using Google Forms. The survey was shared among university students, families, pet owners, and animal lovers to gather insights about the challenges people face while adopting pets and using pet care services. The responses helped in shaping the development of *PetAdopt,* ensuring the application addresses real user concerns and provides a practical, user-friendly experience.

3.2. Survey Objectives

The primary objectives of the survey were:

- To understand the attitudes and behaviour of users toward pet adoption.
- To identify the main challenges faced during the pet adoption process.
- To assess the level of trust users have in existing adoption systems and online platforms.

3.3. Survey Questions

3.3.1 Adopted pet

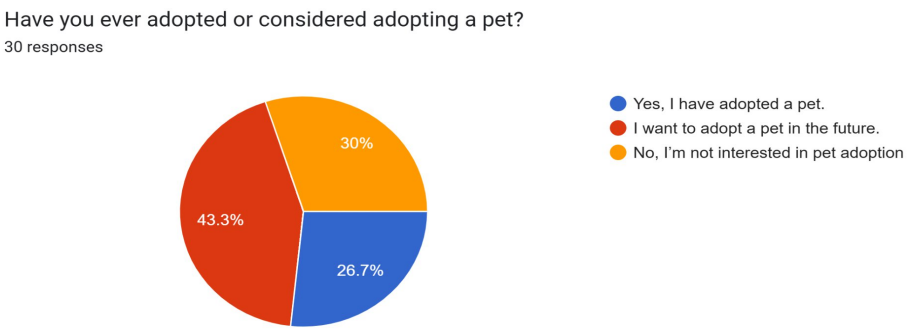


Figure 3.1

The pie chart shows that 26.7% of users already own a pet, 30% are not interested in having one, and 43.3% are considering adopting a pet.

3.3.2 Pet Availability

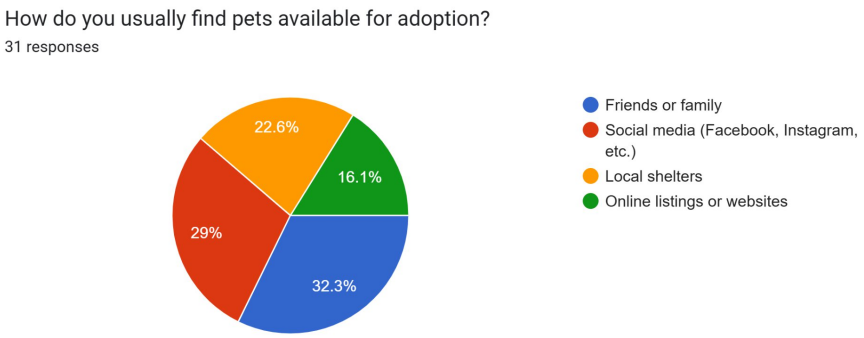


Figure 3.2

The pie chart shows that most people find pets for adoption through friends and family (32.3%), followed by social media (29%), local shelters (22.6%), and online listings (16.1%).

3.3.3 Challenges while adopting a pet

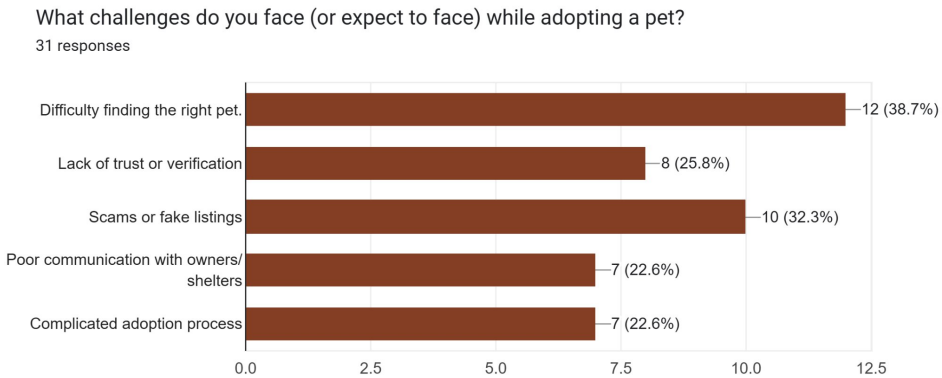


Figure 3.3

The horizontal bar chart shows that the main challenges in pet adoption are scams or fake listings (10 responses) and difficulty finding the right pet (12). Other issues include lack of trust (8), poor communication (7), and complicated processes (7), highlighting major concerns around trust and reliability.

3.3.4 Mobile app Availability

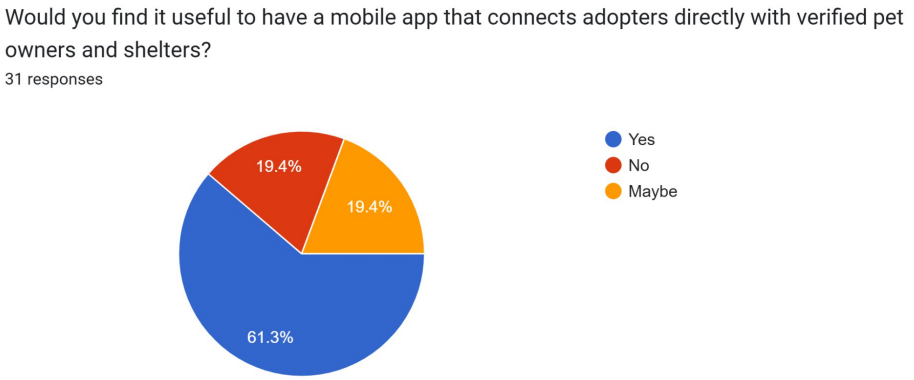


Figure 3.4

Most respondents (61.3%) said yes to wanting a mobile app for pet adoption, while 19.4% said no and 19.4% were unsure, indicating strong interest in a digital solution for easier pet adoption.

3.3.5 Helpful Features

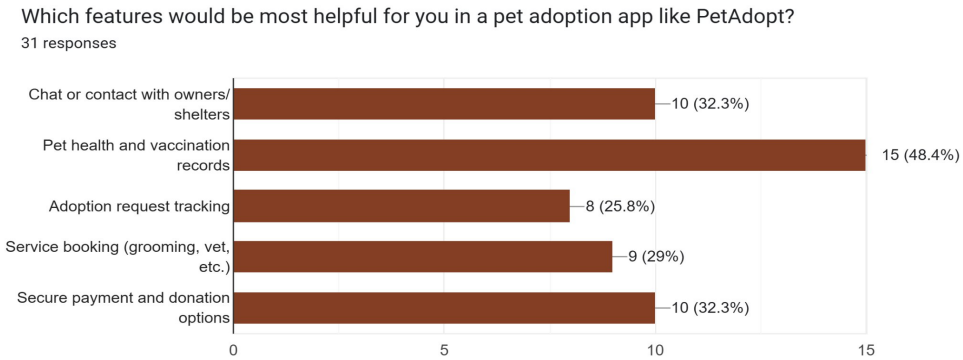


Figure 3.5

The horizontal bar chart results show that the most desired app feature is pet health and vaccination records (15 responses). Other popular features include chat or contact

options (10), secure payments and donations (10), service booking (9), and adoption request tracking (8).

3.3.6 Interest Pet Care Services

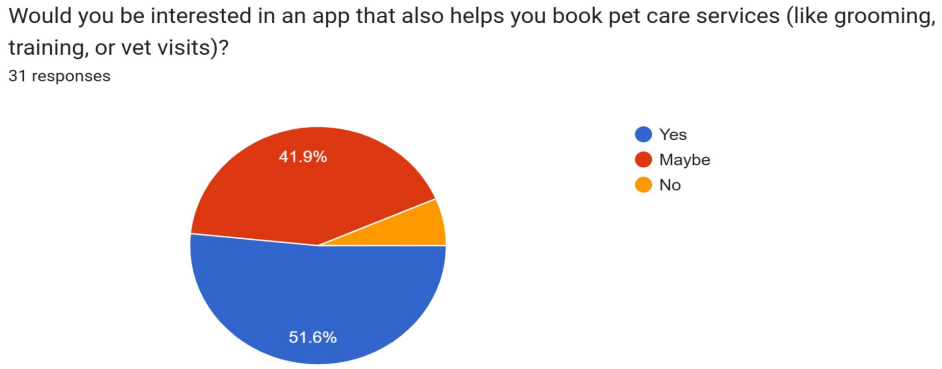


Figure 3.6

The pie chart shows that 51.6% of respondents are interested in pet care services, while 41.9% said *maybe* and the remaining were *not interested*. This indicates a strong potential demand for additional pet care features alongside adoption options.

3.3.7 Likely to use the app

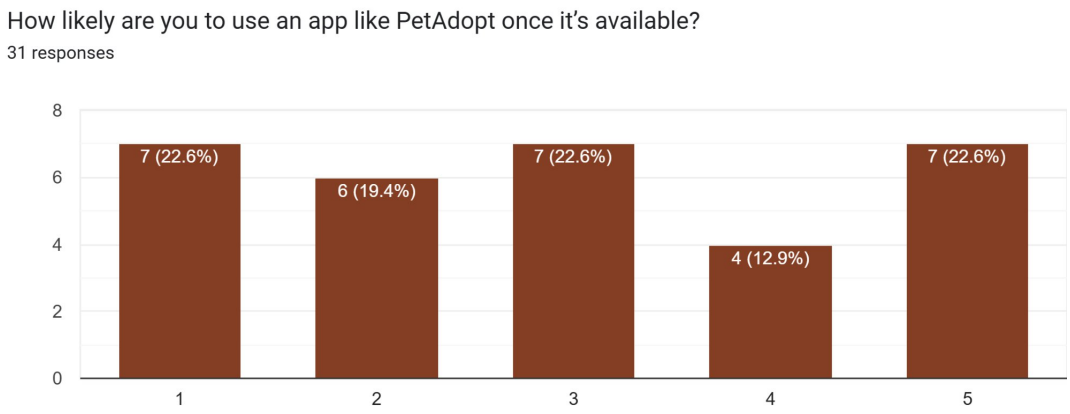


Figure 3.7

The vertical bar chart shows users' likelihood to use the app, rated from 1 to 5. Ratings 1, 3, and 5 each received 7 responses, 2 got 6, and 4 had 4. Overall, most users rated 3 or above, showing generally positive interest in using the app.

3.3.8 Trust

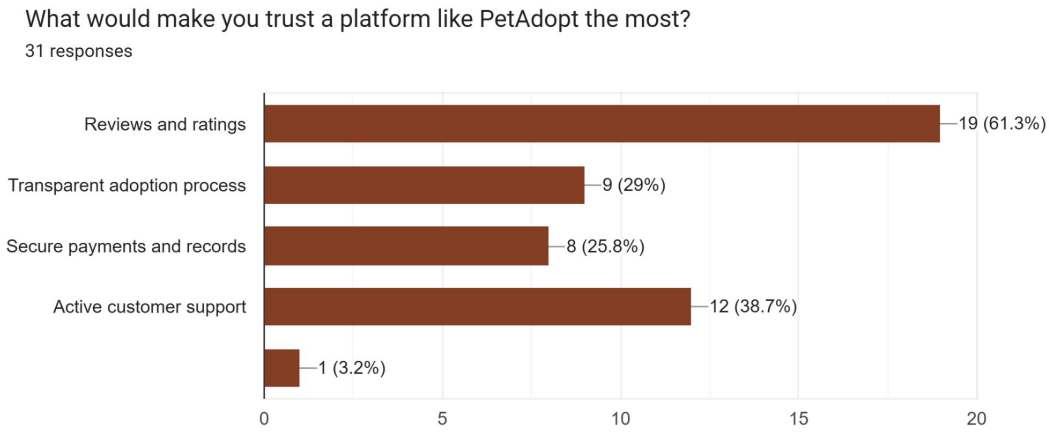


Figure 3.8

The horizontal bar chart shows that reviews and ratings build the most trust (19 responses), followed by active customer support (12), a transparent adoption process (9), and secure payments and records (8). Only one respondent mentioned *other* factors, highlighting that credibility and support are key to user trust.

CHAPTER # 4

SOFTWARE DEMONSTRATION AND IMPLEMENTATION

4.1 Overview

This chapter presents the features and system requirements of PetAdopt to explain the overall functionality of the application. Additionally, a software demonstration is provided to showcase the platform's main operations from the perspectives of Adopters, Shelters/Owners, Service Providers, and Admins. The demonstration highlights essential workflows such as user registration, pet listing, adoption requests, service booking, verification, and approval processes, and real-time communication among different users

4.2 Key Features of PetAdopt

4.2.1 Features of Adopters

- User Registration and Login
- Browse Available Pets
- Search and Filter Options
- Submit Adoption Requests
- Book Appointments or Visits
- Book Pet Care Services
- In App Messaging

4.2.2 Features of Shelters / Pet Owners

- Registration and Verification
- Add and Manage Pet Listings
- Update Pet Availability
- Manage Adoption Applications
- Schedule and Manage Appointments
- In App Communication
- View Adoption History

4.2.3 Features of Service Providers

- Registration and login
- Add and Manage Services
- Appointment Scheduling
- Client Communication
- View Service History
- Receive Ratings and Reviews

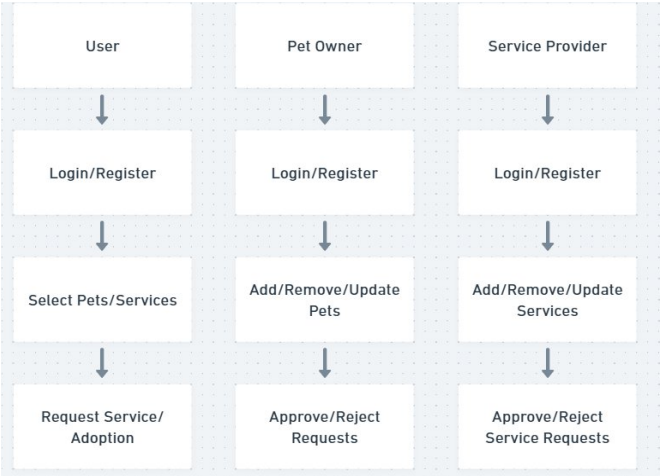


Figure 4.1 Overall Application Flows

Figure 4.2 Pet Owner flow

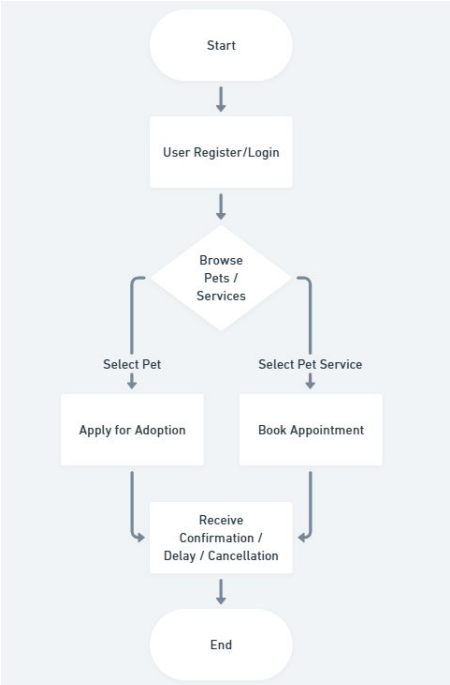


Figure 4.3 Adopter flow (user)

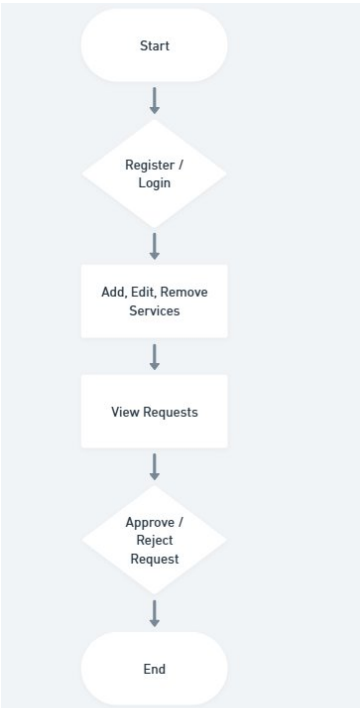


Figure 4.4 Service Provider Flow

4.3 System Requirements

One distinctive kind of system requirement that needs to be met in the course of the development process is software, as well as hardware requirements. The details are as follows:

4.3.1 Software Requirements

No.	Software	Description
1.	Microsoft Office ● Microsoft Word ● Microsoft PowerPoint	Tools for creating and editing the document and slides.
2.	Web Browsers ● Google Chrome ● Firefox ● Edge	Used to gather information related to the project.
4.	Open Source ● Windows 10 ● Windows 11	Used to develop the project.
5.	Development Tools ● Android Studio ● VS Code ● postman	Used to write code scripts for the development system and testing APIs.
6.	Others ● Figma ● Cloudinary ● Scalingo	Prototype, wireframes, storing images, and backend deployment.

Table 4.1 Software Requirement

4.3.2 Hardware Requirement

No	Hardware	Description
1	Laptop	OS: windows10/11 Function: To develop a Flutter app..
2	Pen Drive/ USB	Function: For Transferring Data
3	Projection Connector	Function: intended for Outputting Generated Content such as Resuscitations and Work Documents.

Table 4.2 Hardware Requirements

4.4 Database Design

In PetAdopt, the MongoDB database is used to handle all related data efficiently. The database has several main collections, and each one is used to store a different type of information.

4.4.1 Database of PerAdopt Application

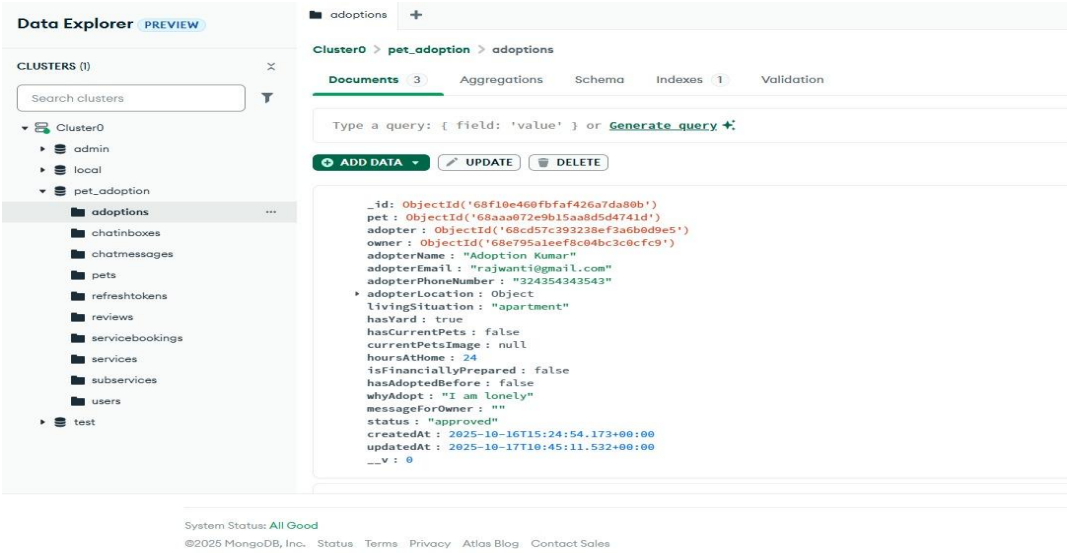


Figure 4.5 Database of Adoption

These are the adoption database and shows detailed information of each appointment.

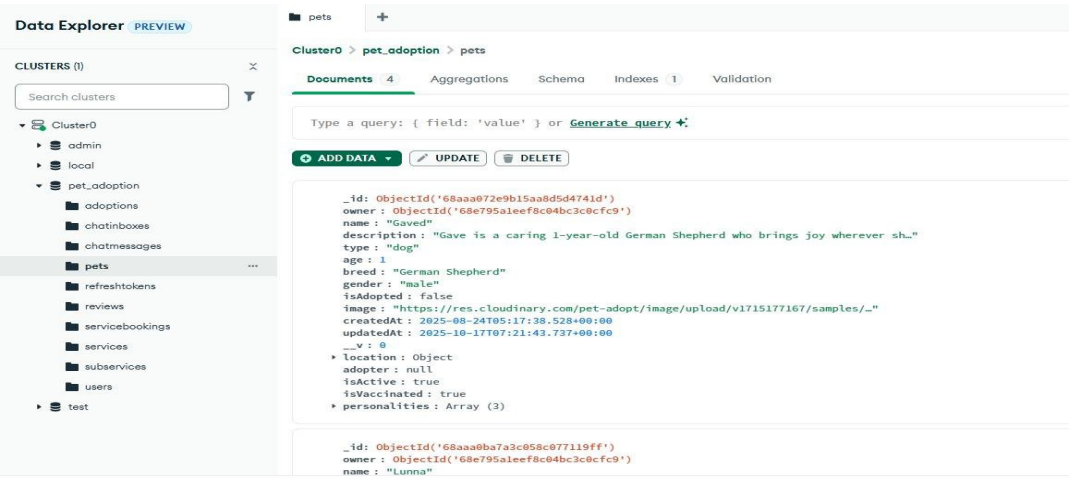


Figure 4.6 Database of pets

These are the pets adopted or available for adoption.

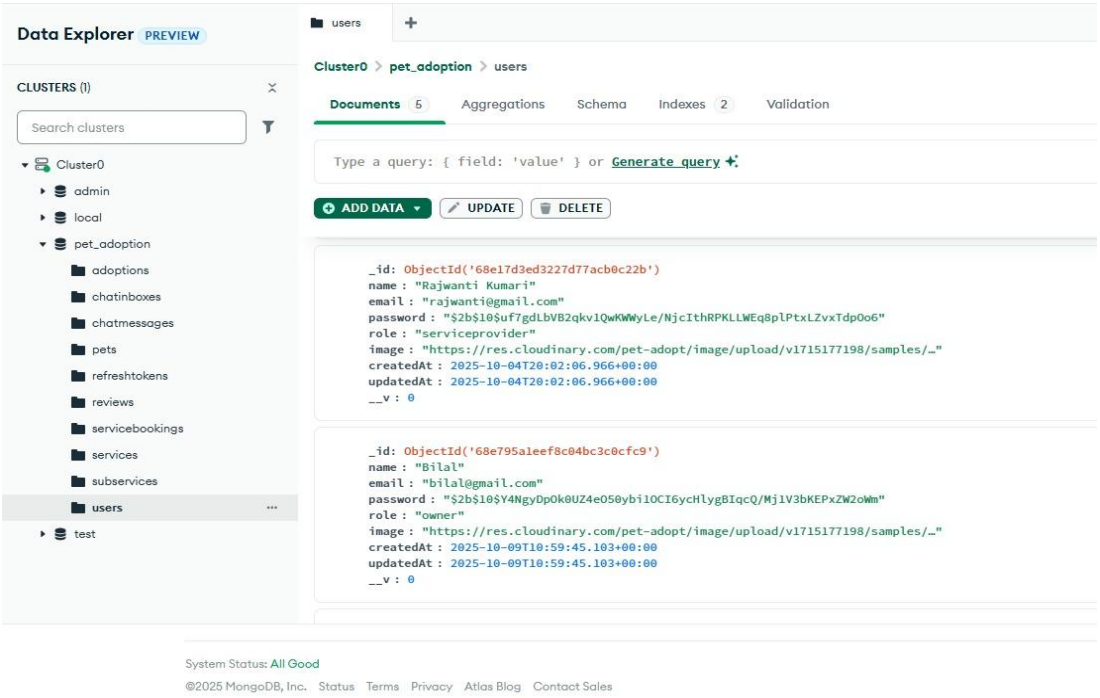


Figure 4.7 Database of users

These are different types of users: pet owners, adopters and service providers.

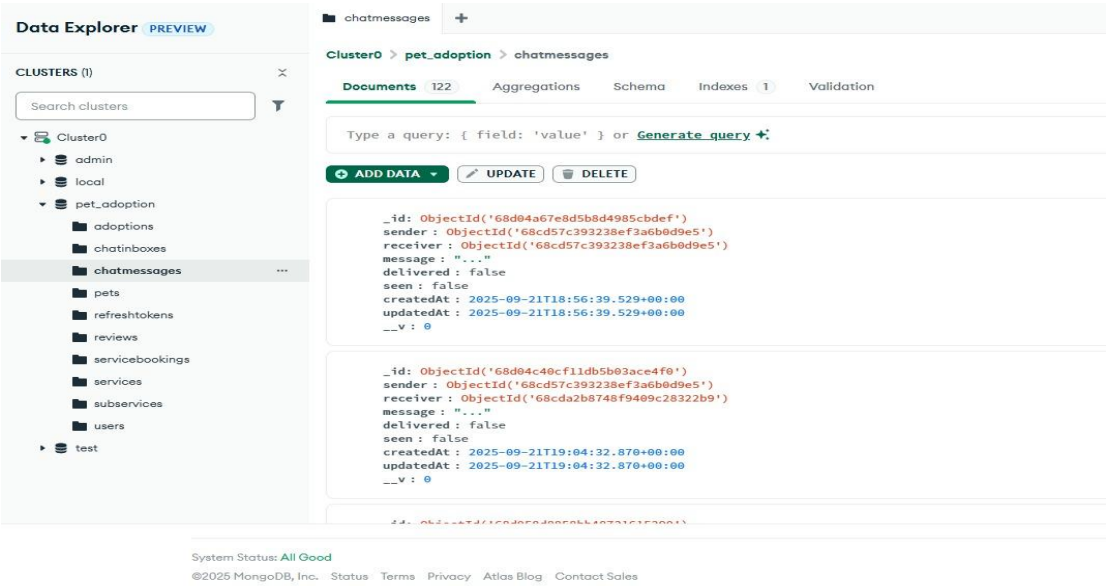


Figure 4.8 Database of messages

These are messages communicated to different types of users.

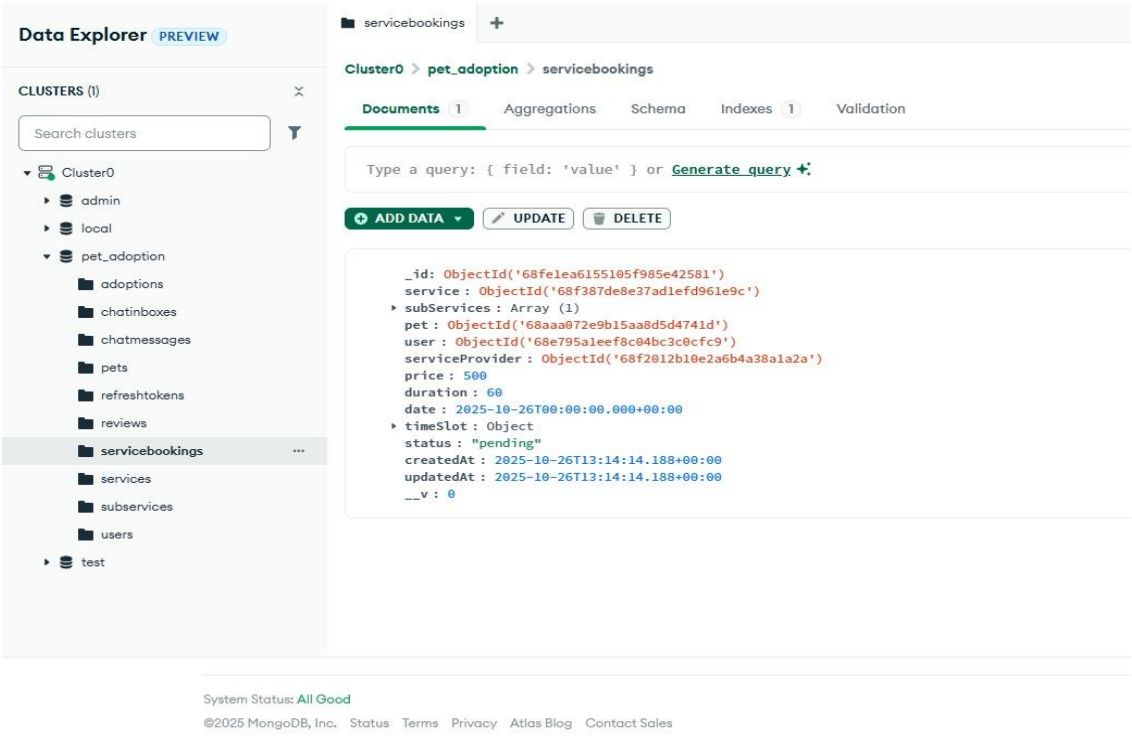


Figure 4.9 Database of Service bookings

These are the data of service bookings done by pet owners and adopters.

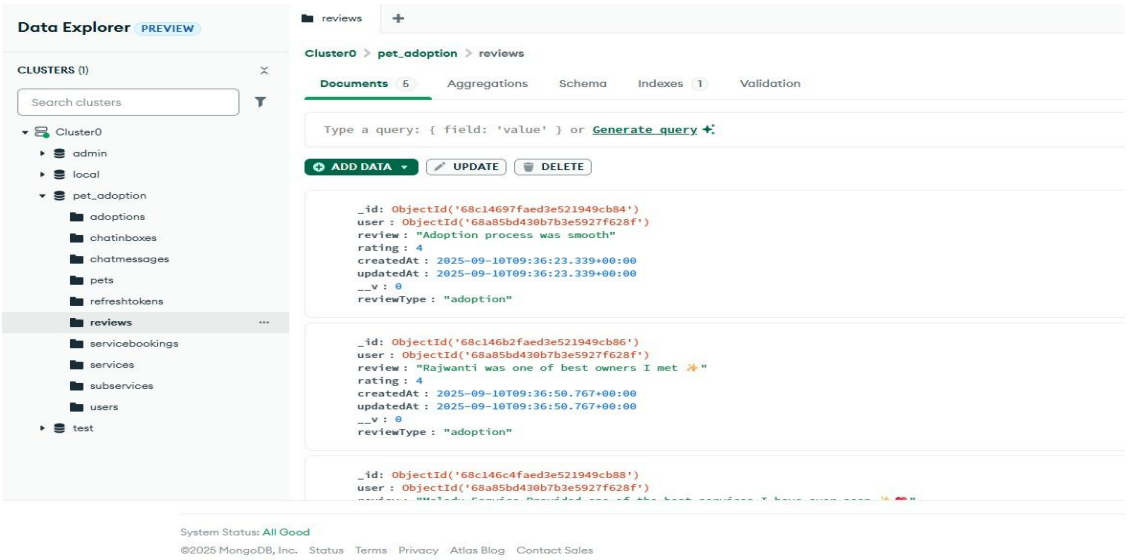
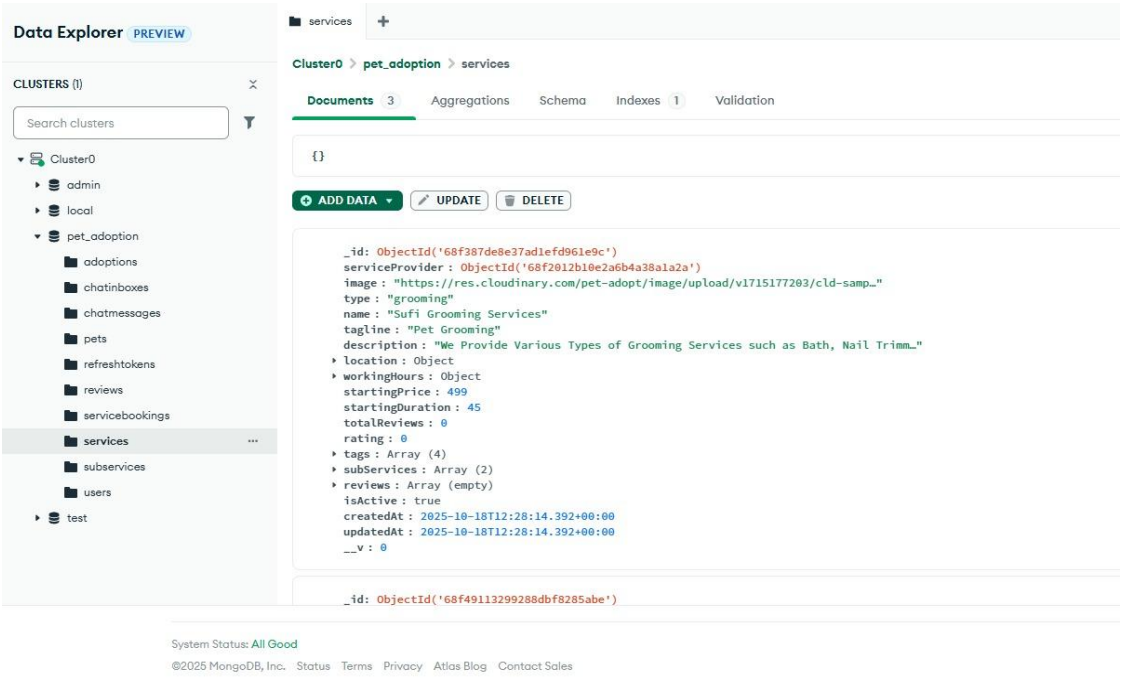


Figure 4.10 Database of reviews

These are databases of reviews given to service providers.



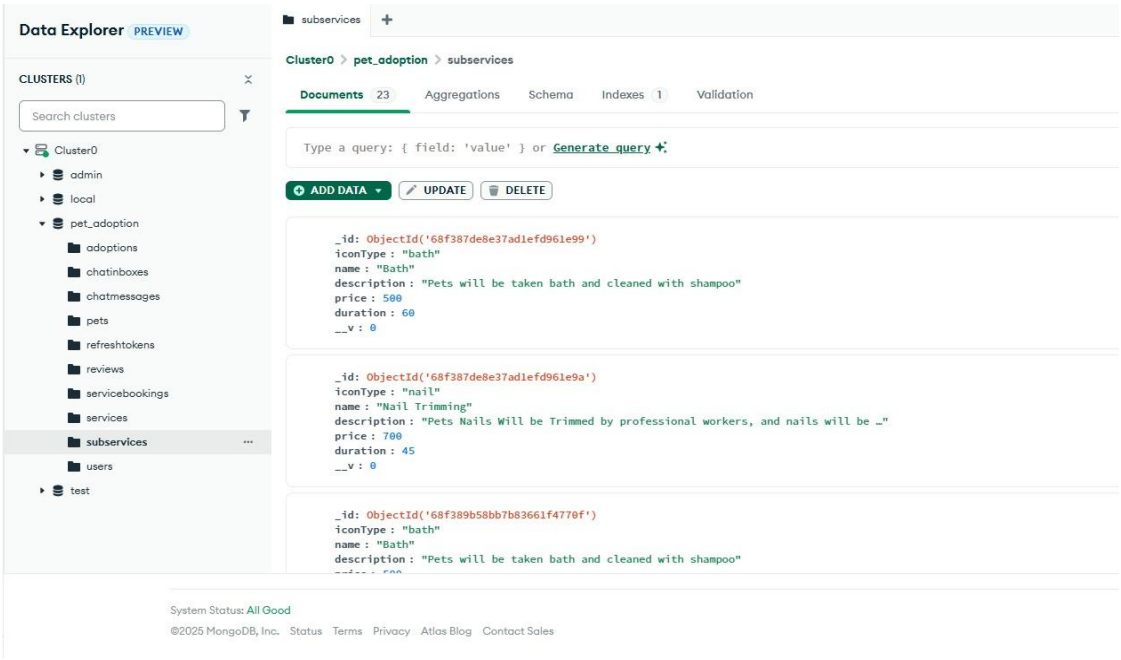


Figure 4.11 Database of services

These are databases of services provided by the service providers.

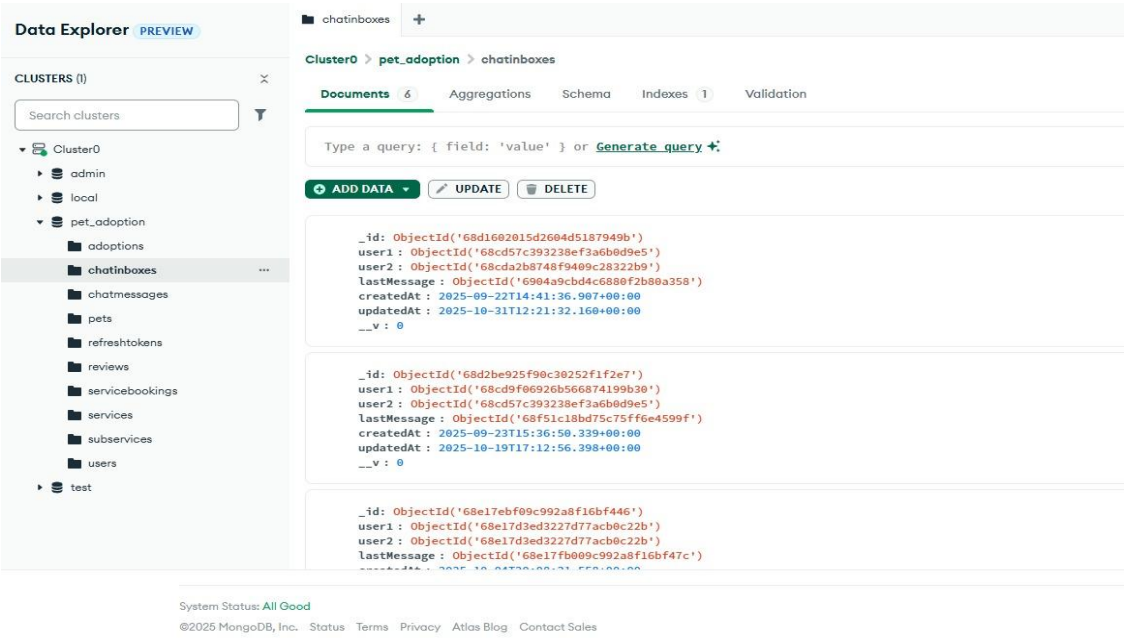


Figure 4.12 Database of chat inboxes

These are databases of chat inboxes, messages that are received but not yet read by users.

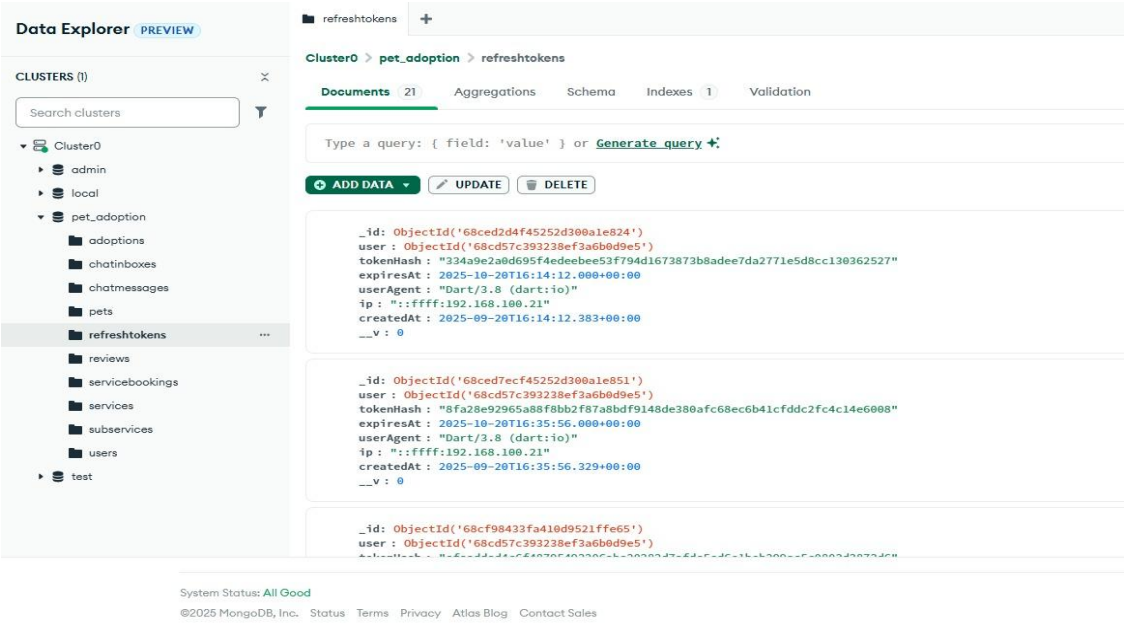


Figure 4.13 Database of Refresh Tokens

These are the database of refreshed tokens that are keeping users logged in.

4.5 PetAdopt Application

4.5.1 Users Screens

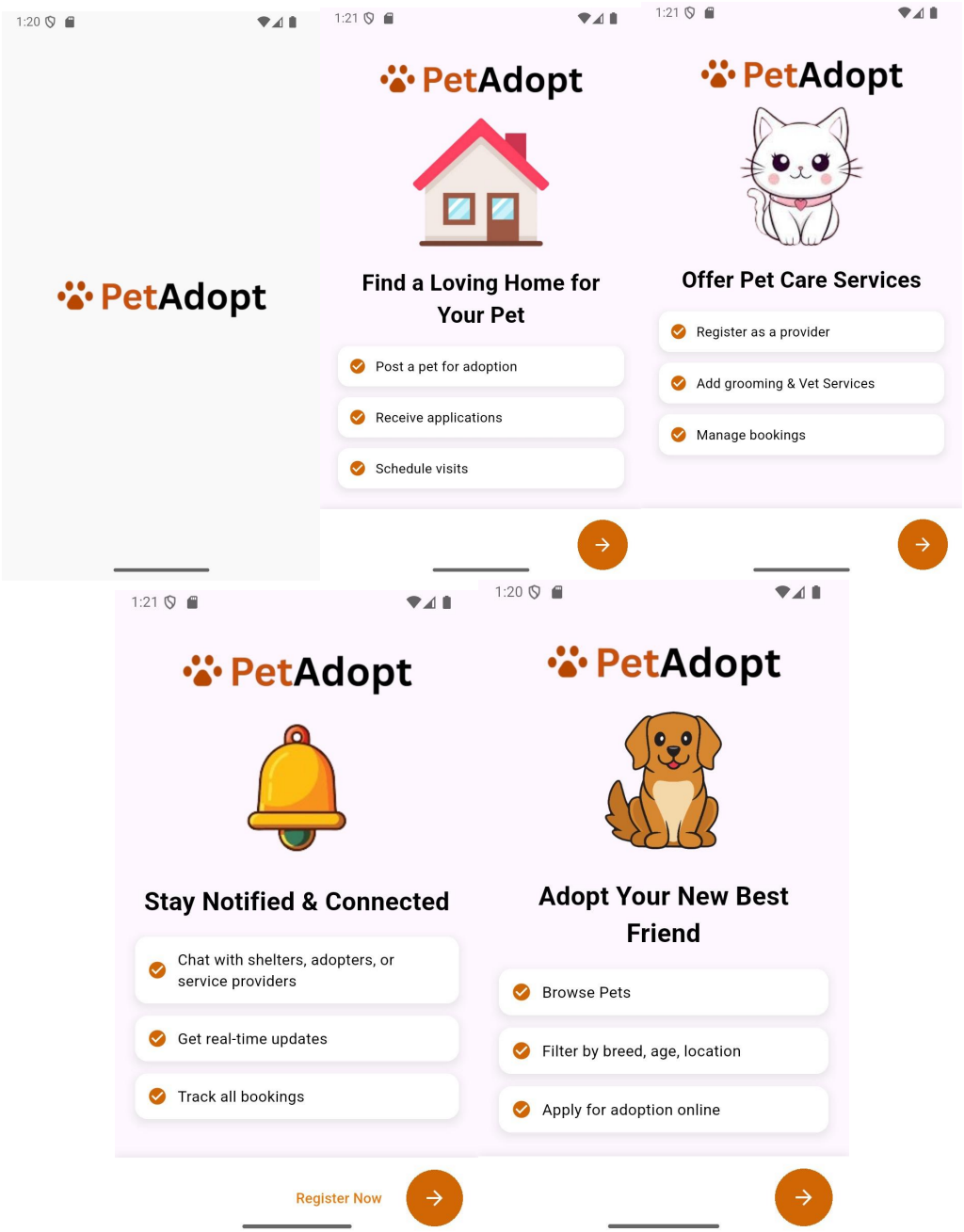


Figure 4.14 Splash screens

From here user can get introductory overview of the project

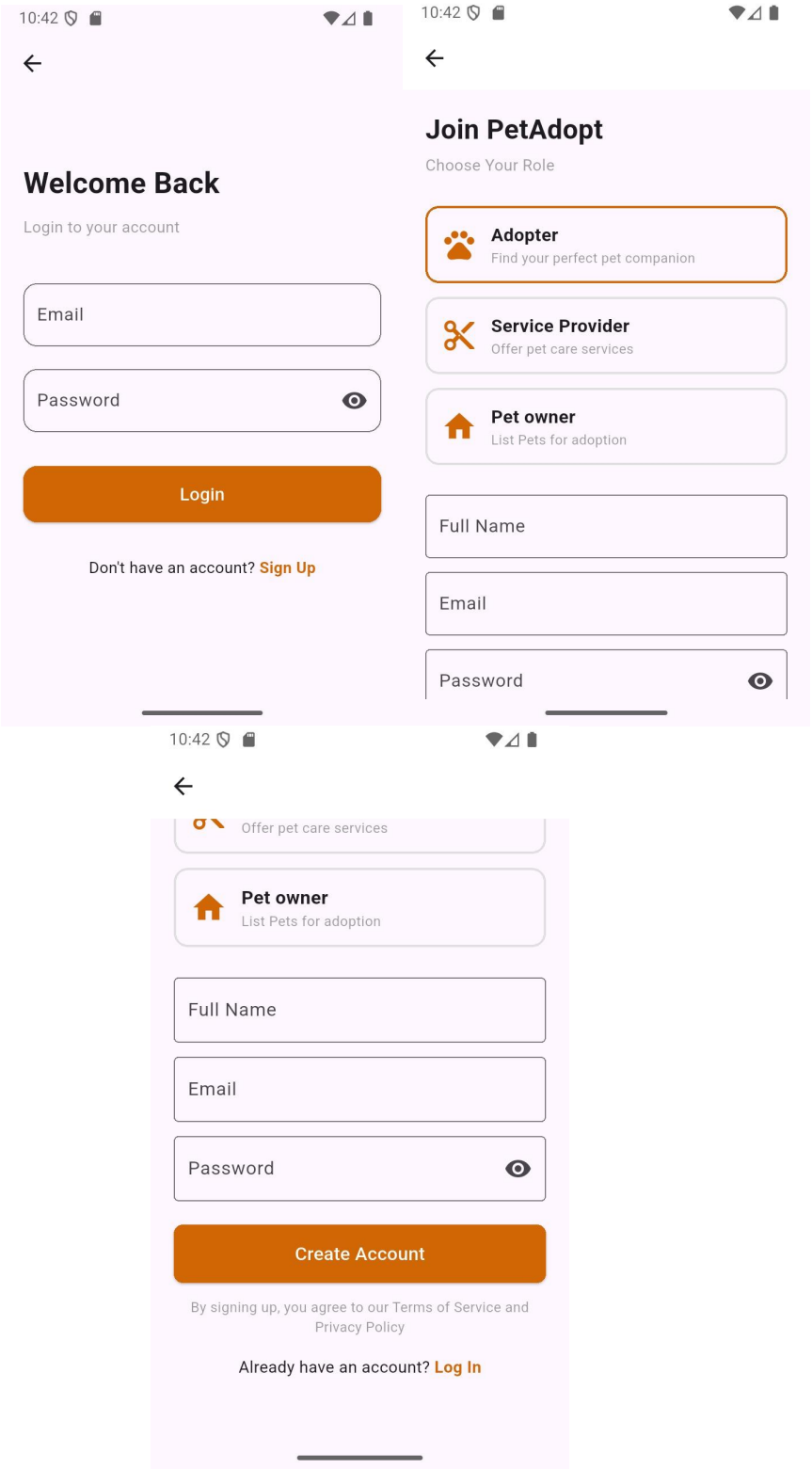


Figure 4.15 Users login/Register Screen

Users need to enter their image and password to SignIn and Signup.

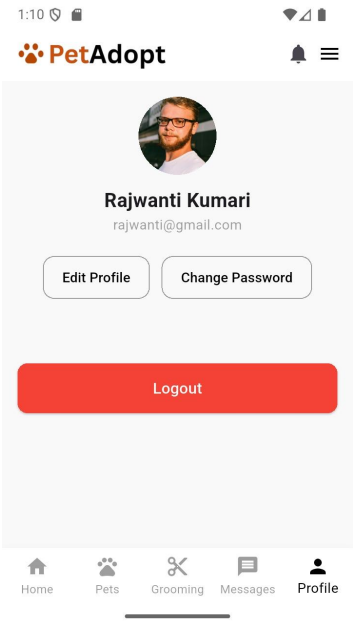


Figure 4.16 User Profile Screen

From here, user can manage their profile setting. It includes My adoptions, Service booked, and Change Password.

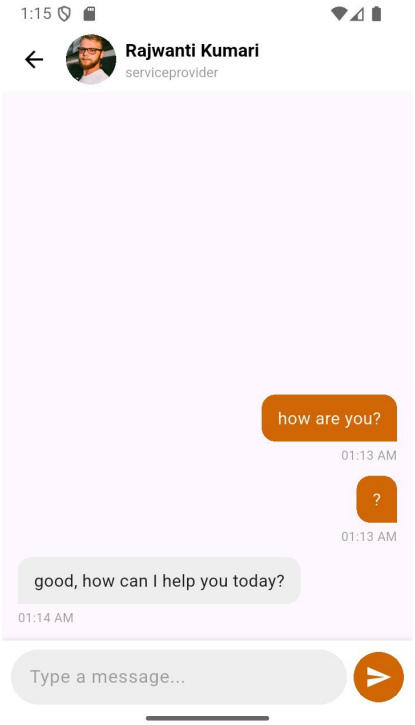


Figure 4.17 User real-time messages

From here, users can real time with each other.

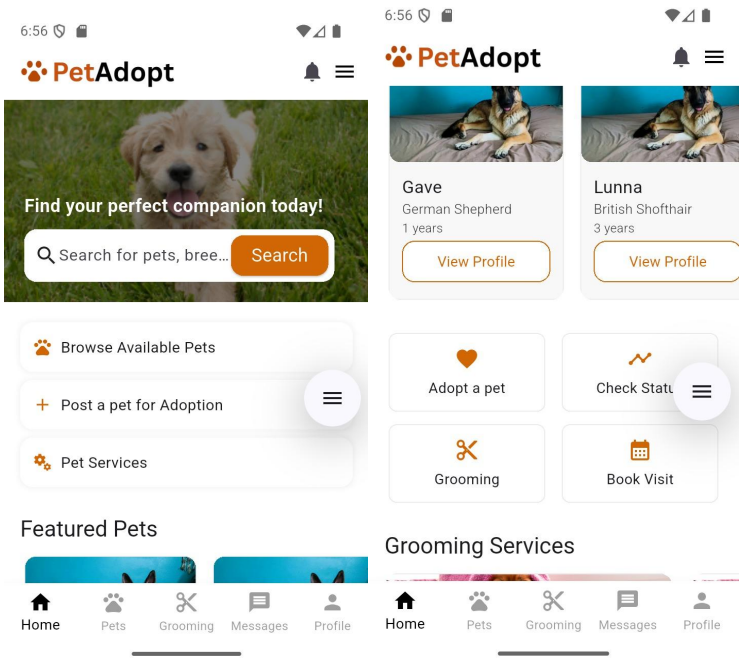


Figure 4.18 Home page

This is the user's screen in which users can browse pets according to their preferences.

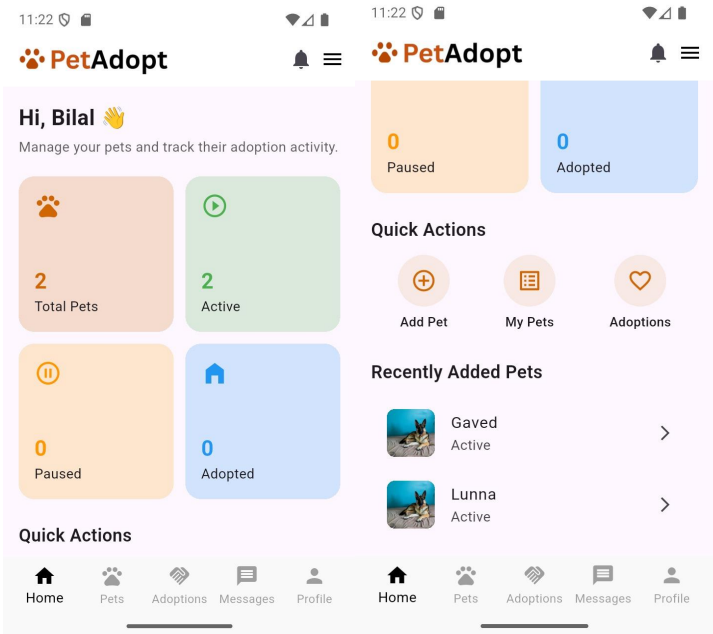


Figure 4.19 Pet Owner Home Page

This is the pet owner's home page screen, in which the pet owner can check statistics related to total pets, adopted pets, and more. The owner can add pets or see recently added pets.

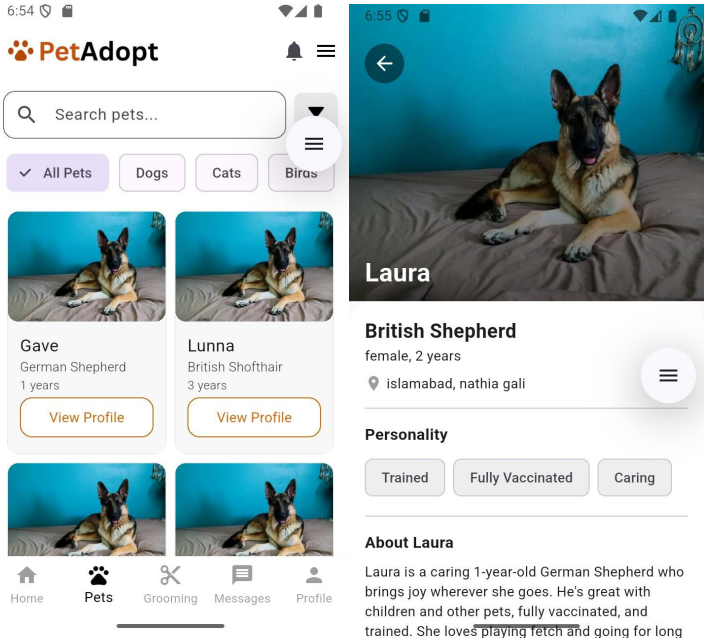


Figure 4.20 Pet Detail page

From here user can get all details about the pet, name, age, location, and personality.

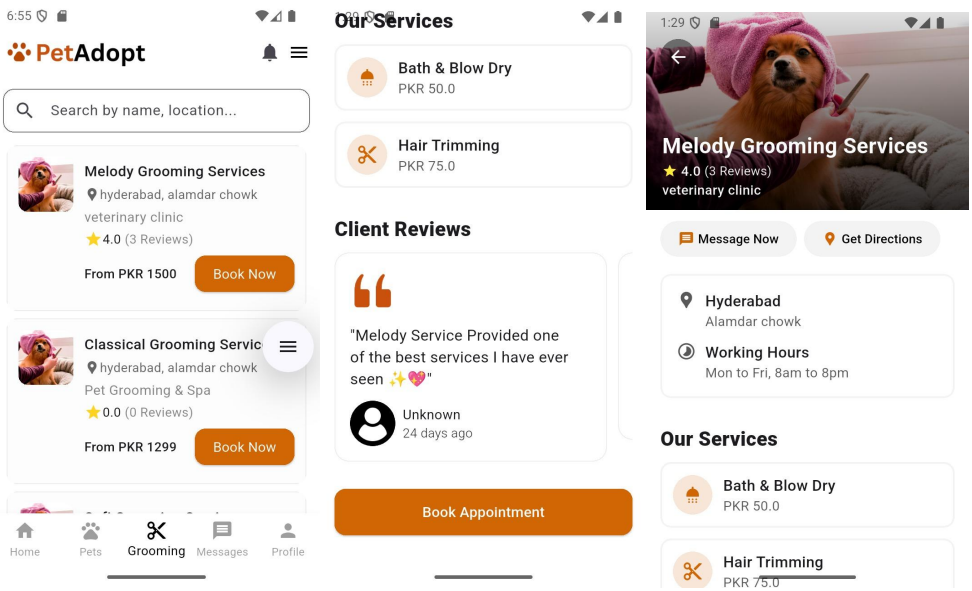


Figure 4.21 Pet service detail page

From here user can get all detail about pet Service includes, including name, location, brand name, reviews, price.

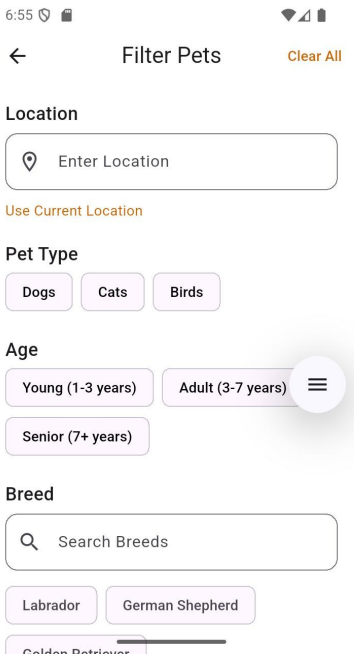


Figure 4.22 Pet Filter Page

From here user can filter pets by pet location, pet type, age, and breed.

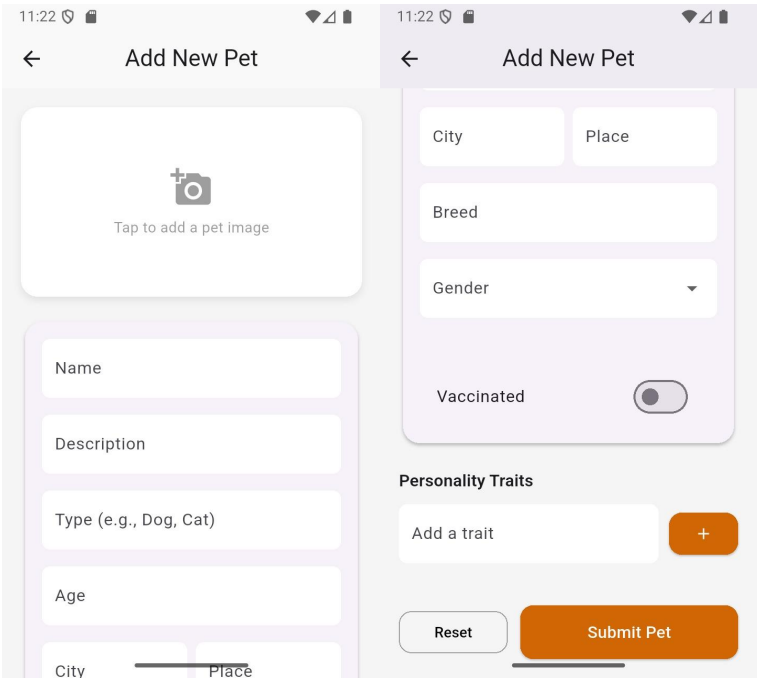


Figure 4.23 Pet Owner Add Pet Page

Owners can add pet information here so they can make their pets available for adoption to users.

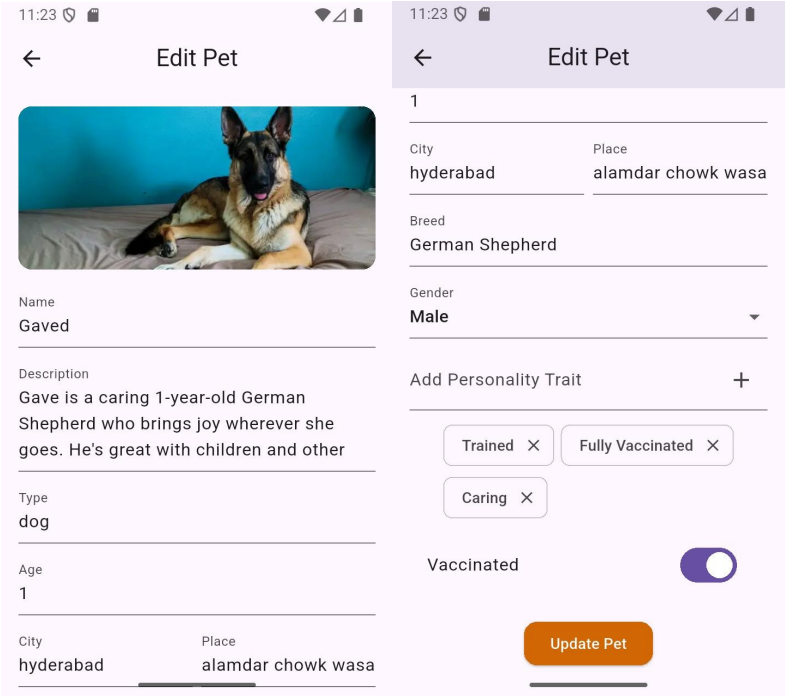


Figure 4.24 Edit Pet Details Page

Owners can edit pet/Update information here.

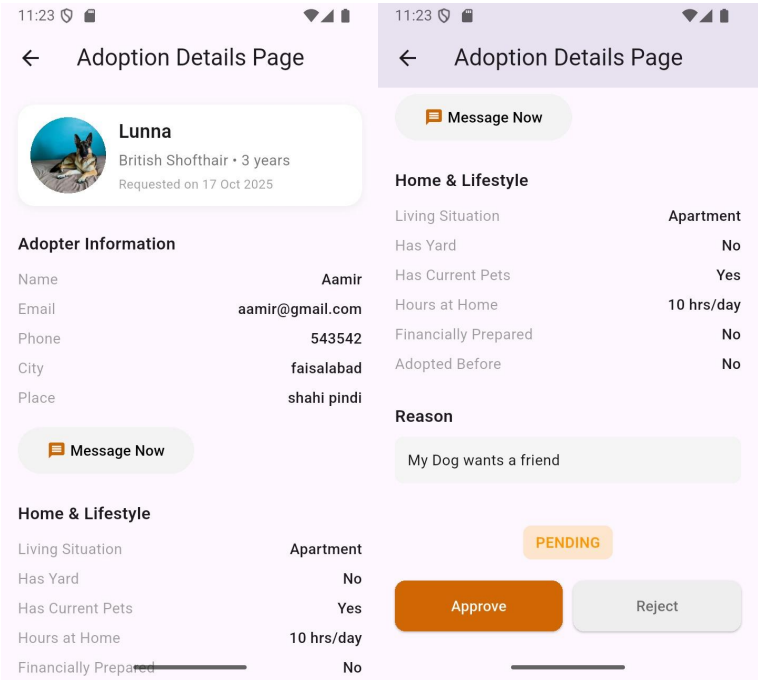


Figure 4.25 Pet Owner Adoption details Page

Here the owner can see details about the adoption filled by the adoptor, and can approve or reject the adoption request.

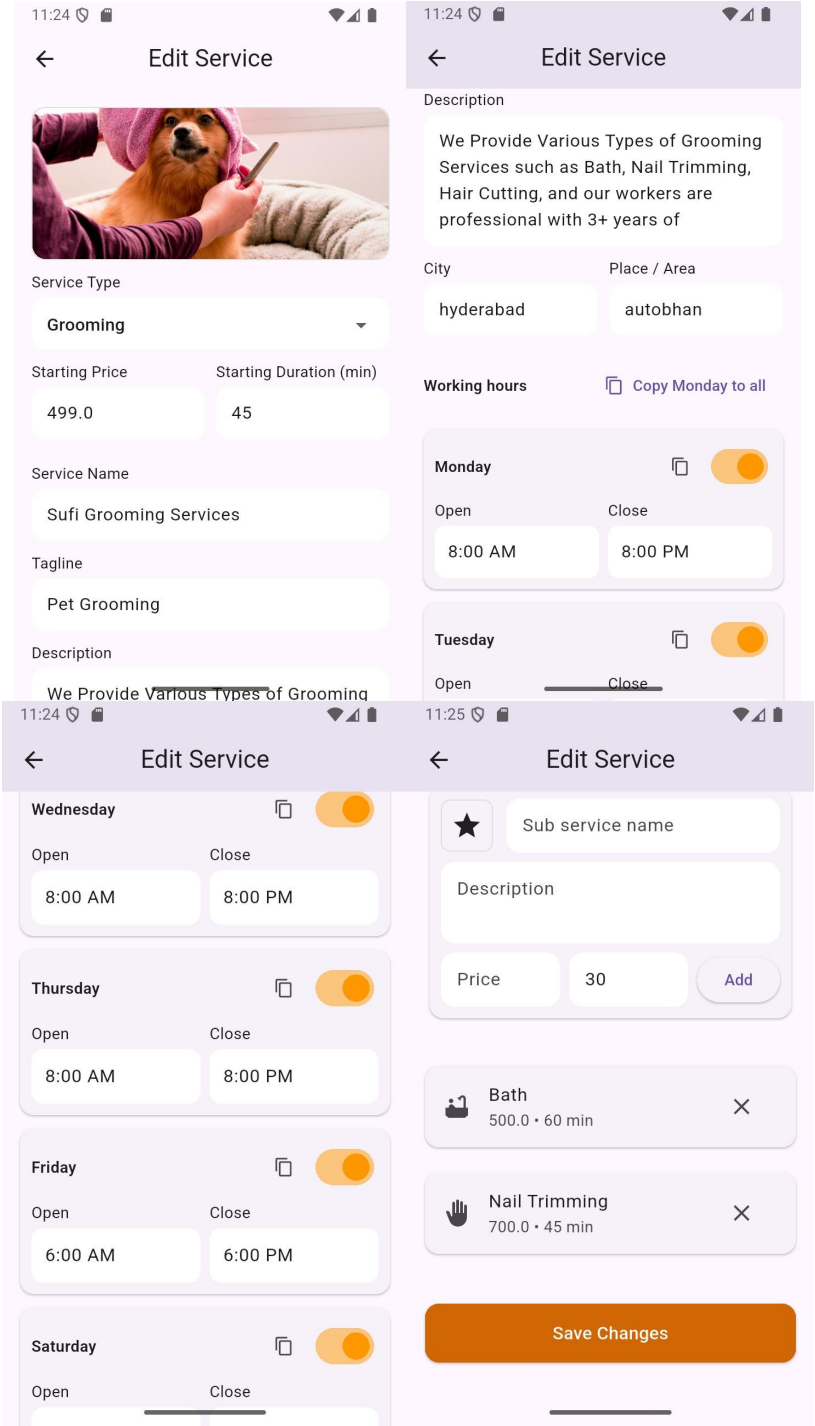


Figure 4.26 Edit Service details Page

Here service provider can edit services.

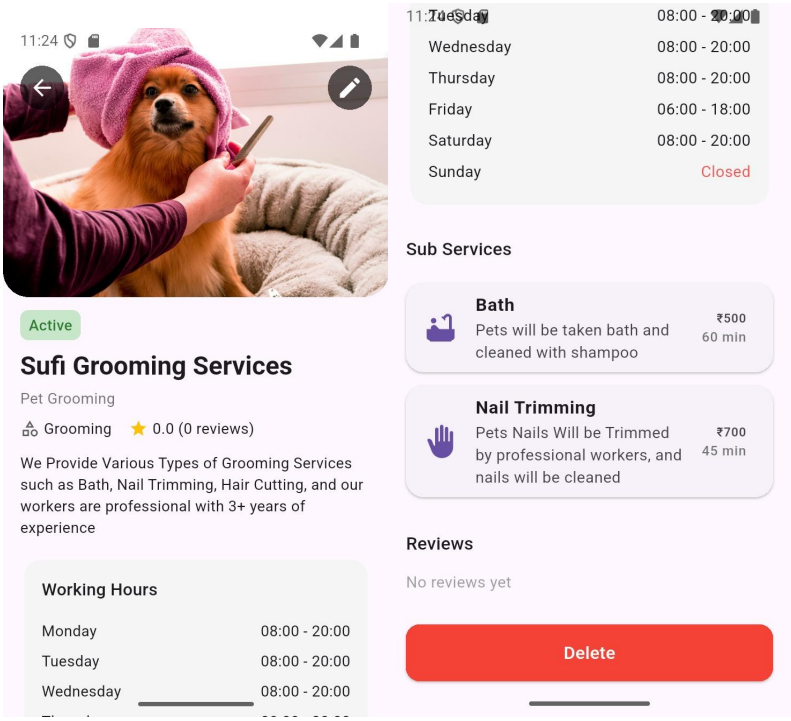


Figure 4.27 Service provider Service Details Page

Here service provider can see details about their service.

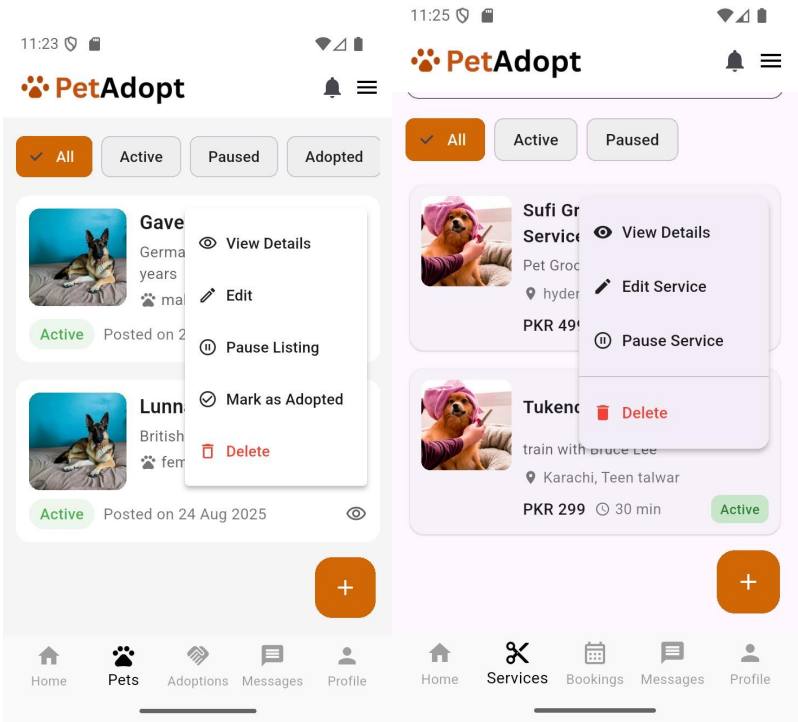


Figure 4.28 Pet owner and service provider pop ups

Pet owners and servicers can views details, edit , changes status and delete listings from easy to navigate pop ups.

CHAPTER # 05
CONCLUSION

5.1 Introduction

This chapter provides an overview of the PetAdopt project's final outcomes, highlighting its impact, achievements, and potential for future growth. The project successfully addressed a key challenge in the pet adoption ecosystem by introducing a reliable, accessible, and user friendly platform that connects pet adopters, owners, and service providers within one digital space. It encourages responsible pet adoption and simplifies the process through technology-driven convenience and transparency.

5.2 Key Contributions

Digital Transformation: PetAdopt digitalizes the traditional and often unorganized pet adoption process, enabling users to explore, adopt, or rehome pets easily through a centralized online platform.

Trust and Safety: The platform promotes secure interactions through verified user profiles, transparent pet details, and admin moderation to ensure ethical adoption practices.

Community Support: Beyond adoption, PetAdopt creates a connected ecosystem by engaging veterinarians, trainers, and groomers making pet care more accessible for adopters and owners alike.

Animal Welfare Impact: By promoting adoption instead of purchasing, PetAdopt contributes to animal welfare awareness, reducing stray populations and promoting compassion in society.

5.3 Limitations

While the system functions effectively, a few areas could be improved in future iterations:

- No current integration with third-party rescue shelters or NGOs.
- Limited personalization features for recommending pets based on adopter preferences.

5.4 Future Work

Several enhancements are planned to expand PetAdopt's reach and usability:

- **In-App Communication:** Secure chat and video call features for better interaction between adopters and pet owners.
- **NGO and Shelter Integration:** Collaboration with verified animal shelters to increase adoption listings.
- **AI-Based Recommendations:** Smart matching system to suggest pets based on adopter lifestyle and preferences.
- **Mobile App Launch:** Android and IOS versions to improve accessibility and adoption-reach.

5.5 Final Thoughts

PetAdopt was envisioned as a social impact solution to promote responsible pet adoption and animal welfare through technology. From research to design and testing, the project proved to be more than a web platform. It's a compassionate ecosystem connecting people and pets in need. With further development, PetAdopt holds strong potential to grow into a nationwide initiative, inspiring kindness and improving lives, one adoption at a time

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